

ZYRQUEN OMEGA ACTIVE ZEROIZATION ATTESTATION

FIPS 140-3 Level 4 Active Tamper Protection & Hardware Recovery Certificate

1. Executive Summary & Test Verdict

A physical tamper fault injection test (Micro-drilling & Tamper Foil Severance) was executed against Utimaco u.trust GP CSe-Series Node TC-03. The Active Zeroization circuit fired instantaneously, shredding all Dilithium-5 private keys in volatile RAM within 0.48 ms (well below the 1.20 ms SLA limit). The system successfully entered Fail-Closed mode, isolated the event to Chamber 02, and performed a seamless Crypto-Agility fallback to SPHINCS+ (FIPS 205) with 0.00% SSoT Baseline Drift.

2. Key Audit Metrics

- Target Node: TC-03 (Dr. Kanyarat Vetchasit / Chief PQC Cryptographer)
- Hardware Standard: FIPS 140-3 Level 4 / CC EAL6+
- Active Zeroization Response Time: 0.48 ms (SLA Limit < 1.20 ms)
- Fallback Signature Scheme: SPHINCS+ (SLH-DSA-192 / FIPS 205)
- Post-Attack Quorum State: 9/10 Signed (Super-Majority >= 8/10 Attained)
- Single Source of Truth (SSoT) Drift: 0.00% (Genesis Merkle Root Immutable)

3. Legal & Statutory Admissibility

Raw tamper logs and volatile state snapshots were automatically cloned and sealed into Module 17 Preservation V24 under strict Zero-Deletion Guarantee. The chain of custody remains 100% admissible under Thai Electronic Transactions Act B.E. 2544 (Sections 9, 26, 28) and PDPA Section 37.

Signed & Certified by: Yutthaphum Phakphian (#EP-SOVEREIGN-01)

Genesis Anchor Block #849202 | Merkle Root: 909ab814479844d8a14816bed34cdbb07528e18501da86fc4691763a43fa4c68